

Zero and Negative Exponents

Answer Key

Grade 8 / Pre-Algebra Expressions and Equations

Quick Answers

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|--------------------|--------------------------|
| 1. 1 | 7. 0.0045 |
| 2. $\frac{1}{25}$ | 8. $\frac{1}{x^2}$ |
| 3. $\frac{9}{4}$ | 9. $\frac{2y^2}{x^3}$ |
| 4. $\frac{3}{x^2}$ | 10. $\frac{1}{5}$, 1, 5 |
| 5. $\frac{1}{x^3}$ | 11. $\frac{1}{16}$ |
| 6. — | 12. $\frac{y^5}{2x^5}$ |
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Curriculum

Level: Grade 8 / Pre-Algebra Expressions and Equations

8.EE.A.1, 8.EE.A.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*

Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

2. *If they got -25: made the answer negative instead of taking the reciprocal*
7. *If they got 4500: moved the decimal point to the right, making the number larger*
11. *If they got -16: read the negative exponent as a negative sign on the value*
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1. Evaluate 7^0 .

Step 1. Use the quotient law on a power divided by itself: $\frac{7^3}{7^3} = 7^{3-3} = 7^0$, and that quotient is plainly 1.

Step 2. So $7^0 = 1$ — and the same argument works for every nonzero base. 1

2. Evaluate 5^{-2} .

Step 1. By definition $a^{-n} = \frac{1}{a^n}$, so $5^{-2} = \frac{1}{5^2}$.

Step 2. $5^2 = 25$, so the value is one twenty-fifth.

$\frac{1}{25}$

Common wrong answer: -25 — the exponent's sign was moved onto the value. The exponent tells you to *reciprocate*, never to negate.

3. Evaluate $\left(\frac{2}{3}\right)^{-2}$.

Step 1. A negative exponent on a fraction flips the fraction: $\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^2$.

Step 2. Square the numerator and the denominator: $3^2 = 9$ and $2^2 = 4$.

$$\boxed{\frac{9}{4}}$$

4. Rewrite $3x^{-2}$ with positive exponents only.

Step 1. The exponent -2 belongs to x alone — the coefficient 3 has no negative exponent, so it does not move.

Step 2. Send only x^{-2} to the denominator.

$$\boxed{\frac{3}{x^2}}$$

Common wrong answer: $\frac{1}{3x^2}$ — the coefficient was dragged down with the variable.

5. Simplify $x^5 \cdot x^{-8}$.

Step 1. Same base multiplied, so the product law applies: add the exponents.

Step 2. $5 + (-8) = -3$, so the result is x^{-3} , which must be rewritten with a positive exponent.

$$\boxed{\frac{1}{x^3}}$$

6. Compare 2^{-3} and 3^{-2} .

Step 1. Neither power can be compared while it is still written with a negative exponent, so evaluate both: $2^{-3} = \frac{1}{8}$ and $3^{-2} = \frac{1}{9}$.

Step 2. Both are unit fractions, and the one with the *smaller* denominator is the larger number: eighths are bigger pieces than ninths.

$$\boxed{\frac{1}{8} > \frac{1}{9}}$$

Teaching note: students who compare the exponents ($-3 < -2$) and stop get this backwards. Evaluate first.

7. Write 4.5×10^{-3} grams as a decimal.

Step 1. $10^{-3} = \frac{1}{1000}$, so the number is $4.5 \div 1000$ — a division, which makes the number *smaller*.

Step 2. Dividing by 1000 moves the decimal point three places to the left: $4.5 \rightarrow 0.45 \rightarrow 0.045 \rightarrow 0.0045$.

$$\boxed{0.0045}$$

Common wrong answer: 4500 — the point was moved the wrong way. A negative power of ten always shrinks the number.

8. Simplify $(x^{-2})^3 \cdot x^4$.

Step 1. Power of a power first: multiply the exponents, $(-2)(3) = -6$, giving x^{-6} .

Step 2. Now the product law: $-6 + 4 = -2$, so the result is x^{-2} .

Step 3. Rewrite with a positive exponent.

$$\frac{1}{x^2}$$

9. Rewrite $\frac{2x^{-3}}{y^{-2}}$ with positive exponents.

Step 1. A factor with a negative exponent changes level: x^{-3} in the numerator moves *down*, and y^{-2} in the denominator moves *up*.

Step 2. Each becomes positive on arrival: $\frac{2x^{-3}}{y^{-2}} = \frac{2y^2}{x^3}$.

$$\frac{2y^2}{x^3}$$

10. Order 5^0 , 5^{-1} and 5^1 from least to greatest.

Step 1. Evaluate each power: $5^0 = 1$, $5^{-1} = \frac{1}{5}$, and $5^1 = 5$.

Step 2. One fifth is less than one, and one is less than five.

$$5^{-1} < 5^0 < 5^1$$

Teaching note: with a base larger than 1, making the exponent smaller makes the value smaller — the negative exponent lands below 1, the zero exponent lands exactly at 1.

11. Find and fix Dana's mistake: Dana wrote $4^{-2} = -16$.

Step 1. Trace the wrong move. Dana computed $4^2 = 16$ and then attached the minus sign from the exponent to the answer, as though 4^{-2} meant $-(4^2)$.

Step 2. The definition says otherwise: the negative exponent sends the power to the denominator, $4^{-2} = \frac{1}{4^2}$.

Step 3. $4^2 = 16$, so the correct value is one sixteenth — a *positive* number smaller than 1, which is the quickest way to see that -16 could not be right.

$$\frac{1}{16}$$

12. Simplify $\frac{3x^{-2}y^4}{6x^3y^{-1}}$.

Step 1. Handle the three parts separately. Coefficients: $\frac{3}{6} = \frac{1}{2}$.

Step 2. The x 's: subtract the exponents, $-2 - 3 = -5$, giving x^{-5} , which belongs in the denominator.

Step 3. The y 's: $4 - (-1) = 5$, giving y^5 in the numerator.

Step 4. Put the three results together.

$$\frac{y^5}{2x^5}$$